

Mikołaj Mołodecki Junior Software Engineer

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Summary

Full-Stack Developer merging web technologies with hardware and multimedia. From building the published PlanPM mobile app (Flutter/Supabase) to programming IoT drone systems and 3D rendering engines, I build end-to-end solutions. I thrive in Agile teams, combining technical problem-solving with a collaborative, team-first mindset.

Skills

Programming

JavaScript/TypeScript, Dart, C#, Python, SQL, C++, Java

Environment

- Version Control: Git, Github, GitLab
- Management Tools: YouTrack, Trello
- CI/CD Pipelines (Github Actions)
- Others: Docker

Web & App Development

- Game dev: Unity, HLSL
- Frontend: HTML, CSS, Tailwind CSS, React
- Backend: Flask, FastAPI, Postman
- Mobile: Flutter, React Native

Databases

MySQL, PostgreSQL, SQLite, Supabase, Appwrite

Professional Experience

06/2025 – 09/2025

Szczecin

Global Logic

Junior software developer intern

Project: IoT Project-Based Learning – World of Drones

- **Frontend Development:** Developed the frontend of the application using **React** and supported the integration of the **FastAPI** backend.
- **AI & Computer Vision:** Developed and deployed an AI model based on Nano architecture for real-time car and license plate detection.
- **Embedded & Hardware Design:** Designed and modeled 3D protective casing for Raspberry Pi and STM32 microcontrollers using **Autodesk Fusion**. Created physical landing pads for autonomous drone operations.
- **System Architecture & Planning:** Participated in the end-to-end system design process, including hardware selection, budgeting, and cost estimation (BOM).
- **Agile Methodology:** Worked in a small, cross-functional team following **Scrum** frameworks, utilizing **Taiga** for task management and **GitLab** for version control and Jenkins for CI/CD workflows.
- **IoT Integration:** Solved complex connectivity issues between edge devices (Raspberry Pi) and backend services.

06/2022 – 07/2022

Szczecin

Altcom S.J., Szczecin

IT Support & Service Intern

- Configured and maintained UNIX servers, NAS storage systems, and local network infrastructure (Routing, DNS, DHCP).
- Provided direct Helpdesk support, diagnosing and resolving complex hardware and operating system issues.

06/2021 – 07/2021
Szczecin

Przedsiębiorstwo Usług Specjalistycznych MVB
Intern

- Assembled large-scale RACK server cabinets and power distribution units based on detailed electrical diagrams.
- Collaborated with the engineering team to review technical documentation and resolve discrepancies.

Selected Projects

05/2025 – Present

PlanPM

Tech Stack: Flutter, Supabase, Dart

- Engineered and maintained a comprehensive mobile application currently available on **Google Play** and the **App Store**, serving a live user base of students and faculty (500+ users).
- Managed the full software development lifecycle (SDLC), including backend architecture, database integration (**Supabase**), and **CI/CD** pipeline configuration.

09/2026 – Present

NavigatorPM

Tech Stack: React, TypeScript, Node.js, Graph Algorithms

- Developing an indoor wayfinding application designed to help incoming students navigate a highly complex, multi-level university campus.
- Implementing a custom routing engine using advanced graph algorithms to calculate optimal, real-time paths between specific classrooms and facilities.
- Building an interactive admin dashboard for dynamic, zero-code map management, ensuring easy scalability and low maintenance overhead.

05/2026

ALEKSY – Offline Voice Assistant

Tech Stack: Python, Whisper

- Architected a privacy-focused, fully offline voice assistant, processing the entire pipeline—from wake-word detection to speech synthesis—locally on-device.
- Optimized neural network inference for low-power hardware, ensuring stable performance without reliance on cloud APIs or external connectivity.
- Designed a modular pipeline architecture that allows for rapid language model updates and component isolation, ideal for critical, high-availability environments.

06/2025 – Present

Maritime Autonomous Systems Simulator

Tech Stack: C#, Unity, HLSL, Physics Simulation

- Engineered a high-fidelity maritime simulation system featuring dynamic ocean surface generation and real-time wave physics, critical for testing autonomous navigation algorithms.
- Implemented advanced shader programs (HLSL) to achieve realistic water rendering while maintaining real-time performance on simulation hardware.
- Collaborated on the integration of autonomous ship controls, focusing on low-latency state feedback and environmental simulation.

Education

10/2023 – Present
Szczecin

Bachelor of Engineering

Maritime University of Szczecin 

- **Specialization:** Multimedia Systems Programming
- **Ongoing Research & Thesis:** "Design and implementation of a system for generating and rendering ocean surfaces and sea waves in the Unity engine." – Focus on real-time water physics, shader programming, and high-performance rendering.

Core Technical Training (Labs):

- **Software Engineering & Management:** In-depth knowledge of software life cycle, development methodologies (Agile/Scrum), and project management.
- **Multimedia & Game Dev:** Extensive experience in game development and simulators using **Unity**, including advanced theory and real-time visualization.

- **Networking & Security:** Hands-on experience in designing local networks (VLAN, ACL) and fundamentals of **Cryptography**.
- **Data Structures & Algorithms:** Solid foundation in algorithm design and efficient data management.
- **Database Management:** Designing relational structures and optimizing queries.

09/2019 – 04/2023
Szczecin

Technical Education

Technical School of Digital Technologies

- **Specialization:** Internet of Things (IoT)
- **Key Achievement:** Graduated with honors, achieving the highest scores in national professional exams (inf.02, inf.03)
- **Extracurricular Activity – School Radio Station:** Managed technical infrastructure and sound systems for school events. Responsible for daily broadcasts, equipment maintenance, and audio signal routing.

Organisations

04/2025 – Present
Szczecin

IT Science Club

Member

- **Industry Networking & Collaboration:** Represented the Science Club in technical conferences and workshops with leading international companies, including **Kongsberg Maritime**, **Pierce Group**, and firms from the **Automotive** sector.
- **Team Collaboration:** Working in a professional development environment and collaborating with a multi-disciplinary team of developers.

08/2024
Szczecin

EuroSciPy 2024

Volunteer & Participant

- **Active Conference Participation:** Attended technical sessions and workshops focused on the use of Python in scientific computing, data analysis, and engineering.
- **International Networking:** Engaged with global Python community experts and researchers, developing a deeper understanding of the language's ecosystem (NumPy, SciPy, Pandas).
- **Technical & Logistical Support:** Provided full-time organizational support, including technical assistance for speakers and managing infrastructure during the international event.
- **Communication:** Managed coordination tasks and attendee support entirely in English, facilitating a smooth experience for international participants.

Certificates

PL-900: Microsoft Power Platform Fundamentals

Languages

Polish
Native



English
B2/C1



Interests

Sports & Outdoors

Active player in the AZS Volleyball Club at the Maritime University of Szczecin; passionate about skiing and kayaking.

Gaming

Reached Master Tier in League of Legends and Global Top 10,000 inosu!. Currently developing a custom, demanding game from scratch.

Technology

Focused on continuous growth through engineering core gameplay systems, real-time VFX, and advanced rendering techniques.